

LAB 2: Agile Backlog Creation & Sprint Simulation in Jira

Project: Blood Bank Inventory & Emergency Donor Matcher

Name: Bhogala Srika

SRN: PES1UG24AM067

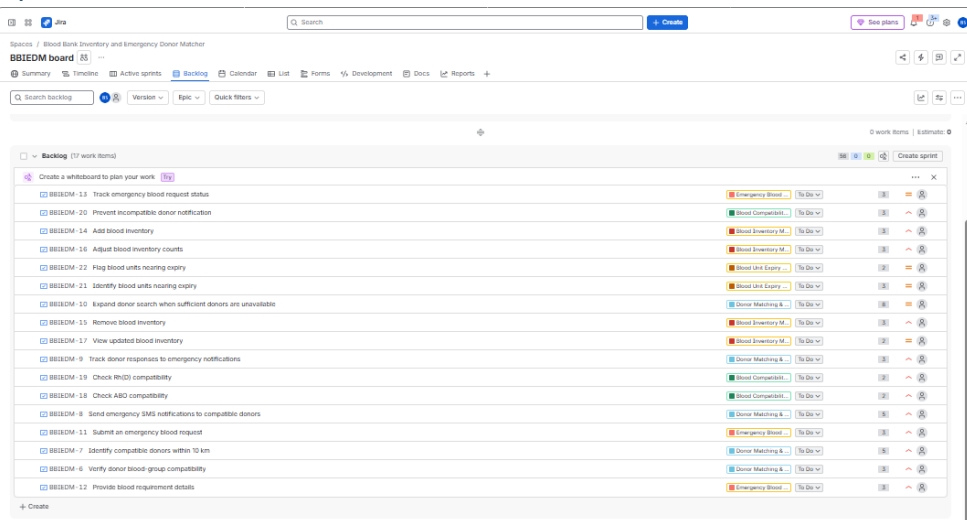
Section: B

Project Overview

The project, **Blood Bank Inventory & Emergency Donor Matcher**, was developed from the functional and non-functional requirements identified in Lab 1. In Lab 2, these requirements were converted into Agile Epics and User Stories, prioritized, estimated using Fibonacci-based Story Points, and organized into two simulated sprints. The User Stories were progressed through the Jira workflow from To Do to In Progress to Done, and sprint progress was monitored using Burndown Charts.

Epic and User Stories

Epic



Donor Matching & Emergency Notification

Description:

Enable the system to identify compatible donors based on blood group and location, notify eligible donors during emergencies, track donor responses, and escalate the search when sufficient donors are unavailable.

☐ > Work	Priority	Status	Story Points	Parent
☐ ☒ Donor Matching & Emergency Notification	Medium	To Do	None	None
☐ ☒ Expand donor search when sufficient donors are un...	Medium	To Do	3	BBIEDM-1 Donor Matching & Emergency Notification
☐ ☒ Track donor responses to emergency notifications	High	To Do	3	BBIEDM-1 Donor Matching & Emergency Notification
☐ ☒ Send emergency SMS notifications to compatible d...	High	To Do	5	BBIEDM-1 Donor Matching & Emergency Notification
☐ ☒ Identify compatible donors within 10 km	High	To Do	5	BBIEDM-1 Donor Matching & Emergency Notification
☐ ☒ Verify donor blood-group compatibility	High	To Do	3	BBIEDM-1 Donor Matching & Emergency Notification
☐ ☒ Emergency Blood Request Management	Medium	To Do	None	None
☐ ☒ Track emergency blood request status	Medium	To Do	3	BBIEDM-2 Emergency Blood Request Management
☐ ☒ Submit an emergency blood request	High	To Do	3	BBIEDM-2 Emergency Blood Request Management
☐ ☒ Provide blood requirement details	High	To Do	3	BBIEDM-2 Emergency Blood Request Management
☐ ☒ Blood Inventory Management	Medium	To Do	None	None
☐ ☒ Add blood inventory	High	To Do	3	BBIEDM-3 Blood Inventory Management
☐ ☒ Adjust blood inventory counts	High	To Do	3	BBIEDM-3 Blood Inventory Management

☐ > Work	Priority	Status	Story Points	Parent
☐ ☒ Provide blood requirement details	High	To Do	3	BBIEDM-2 Emergency Blood Request Management
☐ ☒ Blood Inventory Management	Medium	To Do	None	None
☐ ☒ Add blood inventory	High	To Do	3	BBIEDM-3 Blood Inventory Management
☐ ☒ Adjust blood inventory counts	High	To Do	3	BBIEDM-3 Blood Inventory Management
☐ ☒ Remove blood inventory	High	To Do	3	BBIEDM-3 Blood Inventory Management
☐ ☒ View updated blood inventory	Medium	To Do	3	BBIEDM-3 Blood Inventory Management
☐ ☒ Blood Compatibility Verification	Medium	To Do	None	None
☐ ☒ Prevent incompatible donor notification	High	To Do	3	BBIEDM-4 Blood Compatibility Verification
☐ ☒ Check Rh(D) compatibility	High	To Do	2	BBIEDM-4 Blood Compatibility Verification
☐ ☒ Check ABO compatibility	High	To Do	2	BBIEDM-4 Blood Compatibility Verification
☐ ☒ Blood Unit Expiry Management	Medium	To Do	None	None
☐ ☒ Flag blood units nearing expiry	Medium	To Do	2	BBIEDM-5 Blood Unit Expiry Management
☐ ☒ Identify blood units nearing expiry	Medium	To Do	3	BBIEDM-5 Blood Unit Expiry Management

The User Stories were assigned Fibonacci-based Story Points according to their relative effort, complexity, and uncertainty.

Sprint 1

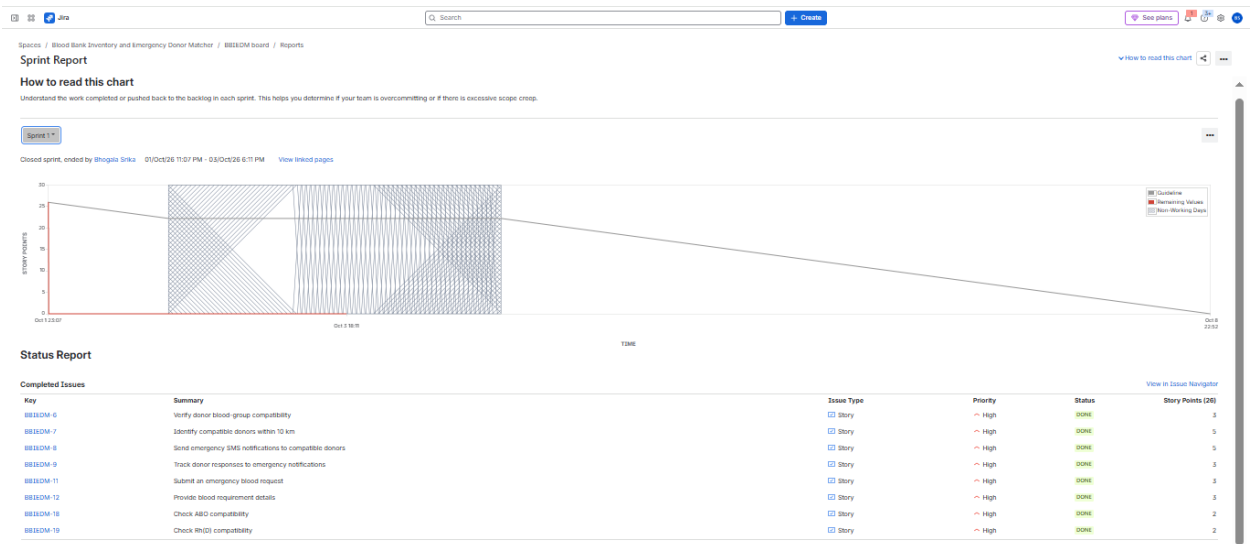
Sprint 1 Scope

User Story	Story Points
Submit an emergency blood request	3
Verify donor blood-group compatibility	3
Identify compatible donors within 10 km	5
Provide blood requirement details	3
Send emergency SMS notifications to compatible donors	5
Check ABO compatibility	2
Track donor responses to emergency notifications	3

User Story	Story Points
Check Rh(D) compatibility	2
Total	26

Sprint 1 was created with 8 User Stories totaling 26 Story Points. The selected stories were progressed through the Jira workflow from To Do to In Progress and finally to Done. After all planned work was completed, Sprint 1 was closed

Sprint 1 Report



Sprint 1 Burndown

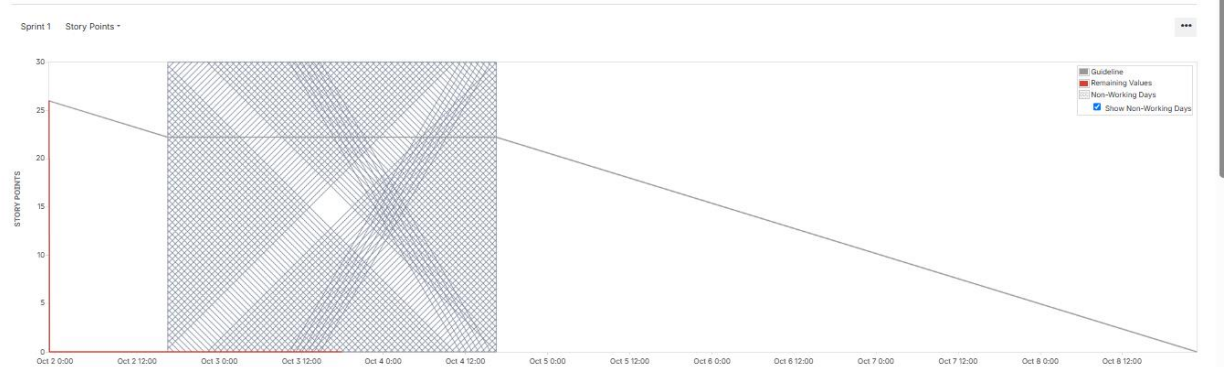
Spaces / Blood Bank Inventory and Emergency Donor Matcher / BBEDM board / Reports

Burndown Chart

How to read this chart

How to read this chart

Track the total work remaining and project the likelihood of achieving the sprint goal. This helps your team manage its progress and respond accordingly.



Sprint 2

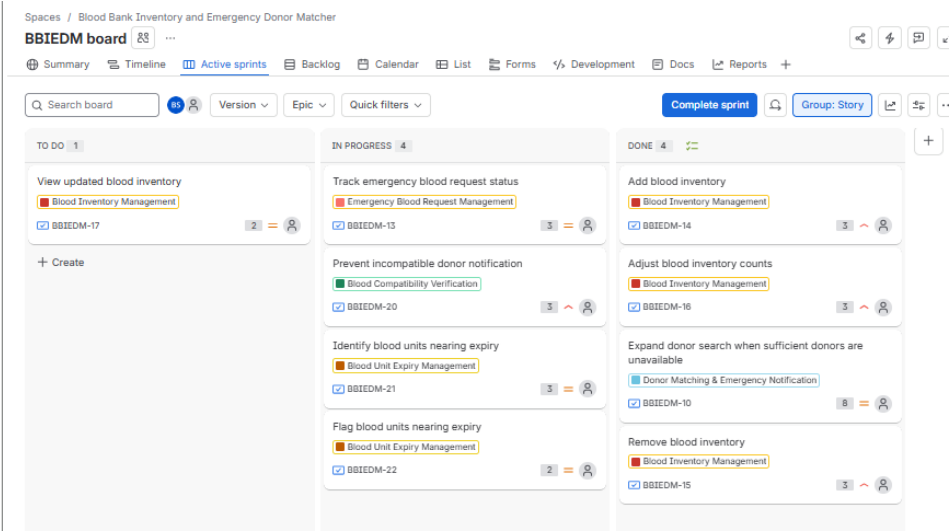
Sprint 2 Scope

User Story	Story Points
Expand donor search when sufficient donors are unavailable	8
Track emergency blood request status	3
Remove blood inventory	3
Add blood inventory	3
Adjust blood inventory counts	3
View updated blood inventory	2
Prevent incompatible donor notification	3
Identify blood units nearing expiry	3
Flag blood units nearing expiry	2
Total	30

Sprint 2 Execution

The remaining 9 User Stories were allocated to Sprint 2, with a total estimated effort of 30 Story Points. The sprint was started and the User Stories were progressed from To Do through In Progress to Done. All planned stories were completed before the sprint was closed

Sprint 2 Active Sprint



Sprint 2 Report



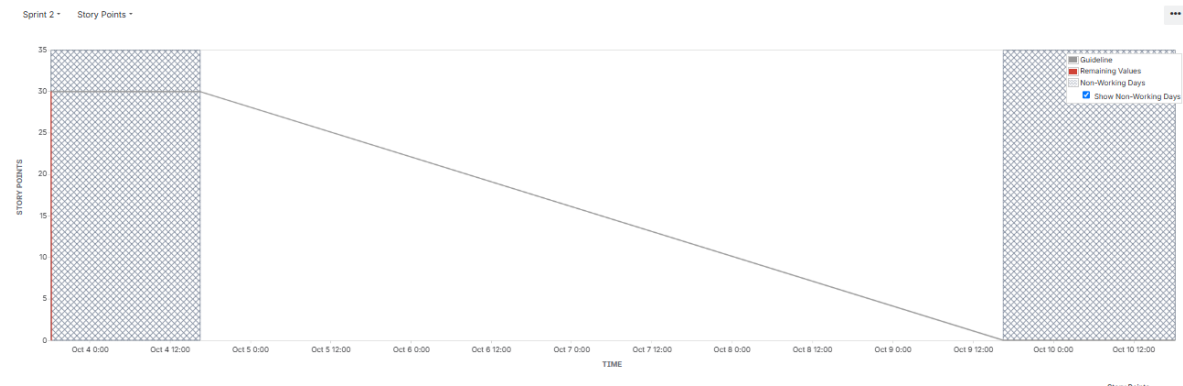
Sprint 2 Burndown

Burndown Chart

[How to read this chart](#)

How to read this chart

Track the total work remaining and project the likelihood of achieving the sprint goal. This helps your team manage its progress and respond accordingly.



Sprint Summary

Sprint	Stories	Story Points	Result
Sprint 1	8	26	Completed
Sprint 2	9	30	Completed
Total	17	56	Completed

Reflection

1. Did your estimations reflect the actual effort?

The Story Point estimates provided a reasonable relative representation of the effort required for the User Stories. Smaller stories such as checking ABO/Rh compatibility were assigned lower points, while more complex functionality such as expanding the donor search was assigned higher points. Both sprints were completed successfully, indicating that the planned workload was manageable for the simulated sprint exercise.

2. Was your backlog well-prioritized?

The backlog was prioritized according to the importance of the functionality to the emergency blood-matching workflow. Core functions such as emergency request submission, blood-group compatibility verification, donor identification, and emergency notification were prioritized

highly because they directly support the main emergency matching process. Inventory and expiry-related functionality was also included to support reliable blood-bank operations.

3. How did your simulated sprint align with your plan?

The simulated sprints aligned with the planned workload. Sprint 1 contained 8 User Stories totaling 26 Story Points, while the remaining 9 User Stories totaling 30 Story Points were completed in Sprint 2. The User Stories were moved through To Do, In Progress, and Done as required by the lab exercise. Both planned sprints were completed successfully.

4. What insights did the Burndown Chart give about your team's capacity?

The Burndown Charts showed that the planned work was reduced to zero by the end of both simulated sprints. Sprint 1 reduced 26 Story Points to zero, while Sprint 2 reduced 30 Story Points to zero. This indicates that the planned workload for each sprint was completed within the simulated sprint period. The charts also provided a visual representation of remaining work and progress toward sprint completion.